Operator Fejér–Riesz with gaps: explicit degree bounds

Exact Mathematica computations for Appendix B (choice q = 1)

Basic interval data and rank-one pencil

Form P, Q, R = Q – P, and Δ = –PQ from the endpoints.

In[209]:=

```
ClearAll["Global`*"];

(*Interval endpoints*)
a = {0, 2};
b = {1, 3};
ell = Length[a];

(*Endpoint polynomials*)
p = Expand[Product[x - a[[j]], {j, 1, ell}]];
q = Expand[Product[x - b[[j]], {j, 1, ell}]];
r = Expand[q - p];
delta = Expand[-p * q];

(*Display the polynomials*)
TableForm[{{"P", p}, {"Q", q}, {"R", r}, {"Delta", delta}},
 TableHeadings → {None, {"Polynomial", "Expression"}}]
```

Out[217]//TableForm=

Polynomial	Expression
P	$-2x+x^2$
Q	$3-4x+x^2$
R	$3-2x$
Delta	$6x-11x^2+6x^3-x^4$

Construct the rank-one pencil T(y) = D – y vv^{T} and its polynomial eigenvector u(x); w(x) = u(x)^{T}u(x) is the new strictly positive multiplier.

In[218]:=
```
(*Construct the rank-one symmetric pencil*)
v = Table[Sqrt[(p /. x → b[[j]]) / (D[q, x] /. x → b[[j]])], {j, 1, ell}];

dmat = DiagonalMatrix[b];

tmat[y_] := dmat - y Outer[Times, v, v];

(*Construct the polynomial eigenvector u(x)*)

u = Table[v[[j]] × PolynomialQuotient[q, x - b[[j]], x], {j, 1, ell}];

(*Strictly positive polynomial w(x)*)

w = Expand[u.u];

(*Display the results*)

MatrixForm[Simplify[tmat[y]]]
MatrixForm[u]
Factor[w]
```

Out[223]//MatrixForm=
$$\begin{pmatrix} 1 - \frac{y}{2} & -\frac{\sqrt{3}\, y}{2} \\ -\frac{\sqrt{3}\, y}{2} & 3 - \frac{3\, y}{2} \end{pmatrix}$$

Out[224]//MatrixForm=
$$\begin{pmatrix} & \frac{-3+x}{\sqrt{2}} \\ \sqrt{\frac{3}{2}} & (-1+x) \end{pmatrix}$$

Out[225]=
$$2\left(3 - 3\,x + x^2\right)$$

In[226]:=
```
MatrixForm[u]
```

Out[226]//MatrixForm=
$$\begin{pmatrix} & \frac{-3+x}{\sqrt{2}} \\ \sqrt{\frac{3}{2}} & (-1+x) \end{pmatrix}$$

In[227]:=
```
Collect[Expand[w /. x → s + 3 / 2], s]
```

Out[227]=
$$\frac{3}{2} + 2\,s^2$$

For q = 1, C(y) = R(T(y))$^{-1}$ is affine. The calculation also gives the compressed polynomial eigenvector identity R(x) C(t(x))u(x) = u(x).

In[228]:=
```
(*Construct R(T(y)) and its inverse C(y)*)
rmat[y_] := Sum[Coefficient[r, x, k] × MatrixPower[tmat[y], k], {k, 0, ell - 1}];

cmat[y_] := Simplify[Inverse[rmat[y]]];

(*Display C(y)*)

MatrixForm[FullSimplify[cmat[y]]]

(*Display T(y) C(y)*)

MatrixForm[FullSimplify[tmat[y].cmat[y]]]

(*Compute R(x) C(t(x)) u(x),where t(x)=Q(x)/R(x)*)

t = q / r;

recoveredU = FullSimplify[Cancel /@ (r (cmat[t].u))];

MatrixForm[recoveredU]
```

Out[230]//MatrixForm=

$$\begin{pmatrix} 1-y & \frac{y}{\sqrt{3}} \\ \frac{y}{\sqrt{3}} & \frac{1}{3}(-1-y) \end{pmatrix}$$

Out[231]//MatrixForm=

$$\begin{pmatrix} 1-\frac{3y}{2} & \frac{\sqrt{3}\,y}{2} \\ \frac{\sqrt{3}\,y}{2} & -1-\frac{y}{2} \end{pmatrix}$$

Out[234]//MatrixForm=

$$\begin{pmatrix} & \frac{-3+x}{\sqrt{2}} \\ \sqrt{\frac{3}{2}} & (-1+x) \end{pmatrix}$$

Claim 2: scalar SOS identity and minimal degree

Compute the multipliers $\sigma 1$ and τ in the displayed SOS form, then expand the scalar identity $1 = w\,\sigma 1 + \Delta\,\tau$. The choice $\sigma 0 = \Delta^2\,\tau$ gives $\Delta = \sigma 0 + w\,\Delta\,\sigma 1$.

In[235]:=
```
sigma1 = Expand[1 / 4 ((x - 3 / 2) ^ 2 - 19 / 12) ^ 2 + 1 / 18];
tau = Expand[1 / 2 (x - 3 / 2) ^ 2 + 1 / 24];
sigma0 = Expand[delta ^ 2 tau];
TableForm[{{"sigma1", sigma1}, {"tau", tau}}]
```

Out[238]//TableForm=

sigma1	$\frac{1}{6} - x + \frac{31x^2}{12} - \frac{3x^3}{2} + \frac{x^4}{4}$
tau	$\frac{7}{6} - \frac{3x}{2} + \frac{x^2}{2}$

In[239]:=
```
Expand[w sigma1 + delta tau]
```
Out[239]=
$$1$$

In[240]:=
```
Factor[sigma0]
```
Out[240]=
$$\frac{1}{6}(-3+x)^2(-2+x)^2(-1+x)^2x^2\left(7-9x+3x^2\right)$$

In[241]:=
```
Expand[sigma0 + w delta sigma1]
```
Out[241]=
$$6x-11x^2+6x^3-x^4$$

The four boundary values force $\sigma 1(j) = 1/w(j)$. Interpolate those values with degree at most three to obtain the polynomial used in the minimality argument; its quadratic coefficient is negative.

In[242]:=
```
(*Interpolation of 1/w at the endpoints 0,1,2,3*)
endpointValues = Table[{j, 1 / (w /. x → j)}, {j, 0, 3}];

pInterp = Expand[InterpolatingPolynomial[endpointValues, x]];

TableForm[endpointValues, TableHeadings → {None, {"x", "1/w(x)"}}]

(*Display the interpolating polynomial*)
pInterp
```
Out[244]//TableForm=

x	1/w(x)
0	$\frac{1}{6}$
1	$\frac{1}{2}$
2	$\frac{1}{2}$
3	$\frac{1}{6}$

Out[245]=
$$\frac{1}{6}+\frac{x}{2}-\frac{x^2}{6}$$

Claim 3: scalar certificate for Δ in M(S0)

In[246]:=

```
(*Claim 3:Scalar certificate for Delta in M(S0)*)
(*Natural generators of S0*)
s0 = {1, x, (x - 1) (x - 2), 3 - x};

(*Three nonzero summands in the certificate for Delta*)
deltaTerms = {x ((x - 1) (x - 2) (x - 3)) ^ 2 / 6,
    (x - 1) (x - 2) (x (x - 3)) ^ 2 / 2, (3 - x) (x (x - 1) (x - 2)) ^ 2 / 6};

(*Compute their sum*)
Factor[Total[deltaTerms]]

(*Compute the weighted degrees*)
Exponent[#, x] & /@ deltaTerms

(*Minimum degrees forced by double zeros at endpoints*)

endpoints = {0, 1, 2, 3};

positiveEndpoints =
  Table[Select[endpoints, Function[z, (s0[[j]] /. x → z) > 0]], {j, 1, Length[s0]}];

minSOSDegrees = 2 Length /@ positiveEndpoints;

minWeightedDegrees = minSOSDegrees + (Exponent[#, x] & /@ s0);

TableForm[
 Table[{j - 1, s0[[j]], positiveEndpoints[[j]], minSOSDegrees[[j]], minWeightedDegrees[[j]]},
  {j, 1, Length[s0]}], TableHeadings → {None,
   {"Index", "Generator", "Positive at", "Min SOS degree", "Min weighted degree"}}]

(*At weighted degree at most 6,only the third generator can have
 a nonzero SOS multiplier.Its multiplier must be c x^2 (x-3)^2.*)

remainingTerm = Expand[s0[[3]] x^2 (x - 3) ^ 2];

remainingTerm

Exponent[remainingTerm, x]
```

Out[248]=

$$-((-3+x)\,(-2+x)\,(-1+x)\,x)$$

Out[249]=

$$\{7, 6, 7\}$$

Out[254]//TableForm=

Index	Generator	Positive at	Min SOS degree	Min weighted degree
0	1	0 1 2 3	8	8
1	x	1 2 3	6	7
2	(−2 + x) (−1 + x)	0 3	4	6
3	3 − x	0 1 2	6	7

Out[256]=

$$18\,x^2 - 39\,x^3 + 29\,x^4 - 9\,x^5 + x^6$$

Out[257]=

6

Claim 4: explicit matrix certificate for the isolated point

In[258]:=

```
(*Claim 4:Matrix certificate for the isolated point xi=4*)
(*Compute deltaXi=-Delta(4) and construct G4*)xi = 4;
deltaXi = -(delta /. x → xi);

g4 = {{deltaXi + delta, (x - xi) delta}, {(x - xi) delta, (x - xi)^2 delta}};

deltaXi

MatrixForm[Map[Expand, g4, {2}]]

(*Compute the determinant of G4*)

Factor[Det[g4]]


(*Construct the four terms in the matrix certificate*)

p1 = (x - 1) (x - 2) (x - 3) (x - 4);
p2 = x (x - 3) (x - 4);
p3 = x (x - 1) (x - 2);
p4 = x (x - 1) (x - 2) (x - 3);

(*Scalar SOS multipliers*)

tauPt = x^2 - 3 x + 27 / 8;

rho1 = (4 - x)^2 tauPt / 4;
rho4 = x^2 + 3 / 2 + x^2 tauPt / 4;

(*Polynomial vectors for the rank-one matrix terms*)
```

```
u3 = {-1 / 4, p3 / 6};
u4 = {3 / 4, p4 / 6};

(*Four weighted matrix terms*)

matrixTerms = {x DiagonalMatrix[{rho1, p1^2 / 6}],
    (x - 1) (x - 2) DiagonalMatrix[{0, p2^2 / 2}], 6 (x - 3) (x - 4) Outer[Times, u3, u3],
    (4 - x) (6 Outer[Times, u4, u4] + DiagonalMatrix[{rho4, 0}])};

(*Compute and display the expanded certificate*)

matrixSum = Map[Expand, Total[matrixTerms], {2}];

MatrixForm[matrixSum]

(*Display G4 for comparison*)

MatrixForm[Map[Expand, g4, {2}]]


(*Compute the weighted degrees of the four terms*)

weightedDegrees =
  Table[Max[Exponent[#, x] & /@ Flatten[matrixTerms[[j]]]], {j, 1, Length[matrixTerms]}];

weightedDegrees


(*Express tauPt as a completed square*)

Clear[s];

Expand[tauPt /. x → s + 3 / 2]
```

Out[261]=

$$24$$

Out[262]//MatrixForm=

$$\begin{pmatrix} 24+6x-11x^2+6x^3-x^4 & -24x+50x^2-35x^3+10x^4-x^5 \\ -24x+50x^2-35x^3+10x^4-x^5 & 96x-224x^2+190x^3-75x^4+14x^5-x^6 \end{pmatrix}$$

Out[263]=

$$-24\,(-4+x)^2\,(-3+x)\,(-2+x)\,(-1+x)\,x$$

Out[275]//MatrixForm=

$$\begin{pmatrix} 24+6x-11x^2+6x^3-x^4 & -24x+50x^2-35x^3+10x^4-x^5 \\ -24x+50x^2-35x^3+10x^4-x^5 & 96x-224x^2+190x^3-75x^4+14x^5-x^6 \end{pmatrix}$$

Out[276]//MatrixForm=

$$\begin{pmatrix} 24+6x-11x^2+6x^3-x^4 & -24x+50x^2-35x^3+10x^4-x^5 \\ -24x+50x^2-35x^3+10x^4-x^5 & 96x-224x^2+190x^3-75x^4+14x^5-x^6 \end{pmatrix}$$

Out[278]=

$$\{9, 8, 8, 9\}$$

Out[280]=

$$\frac{9}{8} + s^2$$

Claim 5: obstruction to weighted degree eight

In[281]:=

```
(*Claim 5:Obstruction to weighted degree 8*)(*Natural generators of S1*)
s1 = {1, x, (x - 1) (x - 2), (x - 3) (x - 4), 4 - x};

endpoints1 = {0, 1, 2, 3, 4};

(*Endpoints where each generator is strictly positive*)

positiveEndpoints =
  Table[Select[endpoints1, Function[z, (s1[[j]] /. x → z) > 0]], {j, 1, Length[s1]}];

(*Minimum SOS degrees forced by double zeros*)

minSigma1 = 2 (Length /@ positiveEndpoints);

(*Maximum SOS degrees allowed by weighted degree 8*)

maxSigmaAt8 = 8 - (Exponent[#, x] & /@ s1);

(*Display the degree restrictions*)

TableForm[Transpose[{s1, positiveEndpoints, minSigma1, maxSigmaAt8}], TableHeadings →
  {None, {"Generator", "Positive endpoints", "Min SOS degree", "Max SOS degree at 8"}}]

(*Generators that can have nonzero SOS multipliers*)

survivingGenerators = Pick[s1, MapThread[#1 ≤ #2 &, {minSigma1, maxSigmaAt8}]];

survivingGenerators


(*Polynomial obstruction from the proof*)

Clear[c1, aa, bb, psi, obstruction];

(*General affine polynomial satisfying psi(4)=1*)

psi = 1 + c1 (x - 4);

(*Polynomial identity that would have to hold*)

obstruction = Expand[-psi^2 (x - 4) - aa x (x - 3) (x - 4) - bb x (x - 1) (x - 2)];

(*Display the constant and cubic coefficients*)
```

```
TableForm[{{"Constant coefficient", Factor[Coefficient[obstruction, x, 0]]},
  {"Cubic coefficient", Factor[Coefficient[obstruction, x, 3]]}},
 TableHeadings → {None, {"Coefficient", "Expression"}}]

(*The constant coefficient forces c1=1/4*)

c1Solutions = Solve[Coefficient[obstruction, x, 0] == 0, c1];

c1Solutions

(*Substitute into the cubic coefficient*)

Factor[Coefficient[obstruction, x, 3] /. First[c1Solutions]]
```

Out[286]//TableForm=

Generator	Positive endpoints	Min SOS degree	Max SOS degree at 8
1	0 1 2 3 4	10	8
x	1 2 3 4	8	7
$(-2+x)\ (-1+x)$	0 3 4	6	6
$(-4+x)\ (-3+x)$	0 1 2	6	6
$4-x$	0 1 2 3	8	7

Out[288]=

$$\{(-2+x)\ (-1+x),\ (-4+x)\ (-3+x)\}$$

Out[292]//TableForm=

Coefficient	Expression
Constant coefficient	$4\ (-1+4\,c1)^2$
Cubic coefficient	$-aa-bb-c1^2$

Out[294]=

$$\left\{\left\{c1\to\frac{1}{4}\right\},\ \left\{c1\to\frac{1}{4}\right\}\right\}$$

Out[295]=

$$\frac{1}{16}\ (-1-16\,aa-16\,bb)$$